

3. PCU mode but no power :- a) Overload(overload in display) b) Low battery (low battery in display) c) Short circuit (overload/short-circuit in display)	a) Reduce the load and reset the switch on the front panel (on-off). b) battery has discharged. Recharge the battery after the mains or solar restoration. c) Check or reduce the output load & turn the reset switch on the front panel (OFF-ON). Call electrician to check the short circuit.
4.Backups not satisfactory with PV connected.	a) Connect the PV wire in the correct terminal with correct polarity as per marking . b) The PV panel should be mounted properly with the correct angle. c) Voc should be within the range which is mentioned in the technical specification.

STEPS FOR PCU INSTALLATION

TO BE DONE BY AN AUTHORIZED SERVICE PERSON.

- Switch off the supply to the distribution point to which the PCU unit to be connected.
- For extra safety also remove the fuses from the line. Make absolutely sure with measurement that there is no power.
- A readily accessible disconnect device must be incorporated in all fixed input wiring. The disconnect device shall have a contact separation of at least 3mm. The PCU has automatic back feed isolation but for extra safety a warning label can be added on the primary power isolator installed remote from the PCU to warn electrical maintenance.
- Check the building wiring, improper wiring will not prevent the PCU from operating but will limit its protection capabilities. Improper building wiring could result in equipment damaged that is not covered in warranty.
- Connect the battery/ batteries to PCU as per its requirement.
- Keep the front switch of PCU on OFF Position.
- Measure Voc which is coming from the solar panel, it should be within range which is mentioned in the technical specification.
- Connect solar wires to the system according to marking (Polarity check) on the rear panel of the system.
- Connect the Load wire to the right hole of the output terminal located on the rear panel of PCU.
- Switch on the front switch of the PCU.
- Connect AC input wire to commercial mains terminal.

IMPORTANT

In the event of any instrument requiring service at our authorized service center, the following procedure should be adopted:-

- The instrument must be securely packed, preferably in its original packing. Also ensure that nothing inside packing is damaged. Please transport the product in its original packaging to protect against shock, damage & impact.
- We reserve the right to charge the consignee for any damages incurred during transit.
- The output of the PCU should never be connected to a generator or incoming utility power source.
- This situation is far worse than a short circuit. If the unit survives the condition, it will shut down until correction is made.

DO'S	DON'TS
- Check the battery polarity & connect the product accordingly: - Connect black wire to the -ve terminal of the battery. - Connect red wire to the +ve terminal of the battery. - First connect the battery to the product. Only after that connect the product with solar & mains. - Ensure no reptiles like Mouse, lizards in the PCU room. - Heavy power load should be bypassed from product connection. - Check the solar polarity , recommended Number of solar panels and measure Voc then connect to the product accordingly. - Connect solar PINK wire with (PV+) and BLUE wire with (PV-) on the solar HT terminal. - Fit all connections tightly. - Install the product in covered space with proper ventilation and keep one foot gap between back of product and wall or any obstruction. - Use proper and standard plugs for connections. - Product and battery should be kept in a trolley and dry place. - Always use the recommended battery Ah and number of batteries. - Ensure all vents of the connected battery are not obstructed. - Please contact the authorized service person in case of any issue in the product. - Disconnect Solar , Mains & battery connection before servicing.	- Never connect product to battery without checking correct polarity of battery wires from the product & battery terminals. - Never connect electricity (mains) or Solar without connecting the battery with the product. - The room should not be heated for PCU and battery. - Never connect all the load with the product. - Never connect solar without checking solar VOC. - Never connect the product to the solar without checking the polarity of the solar panel wire and product HT (solar terminal block). - Never leave the loose connections. - Never install the product under the sky, in tight space, near a water tank or kitchen. - Never use loose or nonstandard plugs for connections. - Never place product and battery on floor and dusty. - Never use an un-recommended battery. - Never block the vents of the connected battery. - Do not try to open or repair the product on your own as it may electrocute. - Do not use any liquid or detergent powder for cleaning the product.

TERMS & CONDITIONS

Eastman warrants to the original purchaser provided the product is still in possession and used by the original purchaser. Warranty of 60 months will be provided from the date of sale to customer/dispatch of the instrument if used within its specification. The warranty for the replaced components will lapse along with the warranty of the main instrument. Eastman Auto and Power Ltd. reserves the right to make changes in design and specification without any prior notice and without any obligation to install such changes on units previously supplied. Consumer shall have no coverage or benefit under this warranty in the event any of the following conditions are applicable.

- The product has been subjected to abnormal use or conditions,

improper storage, exposure to excessive moisture or dampness, exposure to excessive temperature, unauthorized modification, unauthorized repair(including but not limited to the use of unauthorized spare parts in repairs), abuse, accident, act of God, spill of food or liquid, improper installation and breakage or damage.

2. The consumer has not promptly notified and lodged a complaint regarding the defect of the product to the company during the warranty period.

3. The product Serial No. code or the accessory code has been removed, defaced, or altered.

4. The seals are broken or show evidence of tampering.

5. Any plastic surfaces and/or any other externally exposed parts are scratched or damaged due to abnormal customer use.

6. The problem of Miniature Circuit Breaker/External Short Circuit blown will not be included in the warranty of the product. The service given for the same will be a paid service.

Plastic components like cables, input/output terminal, battery connector etc. will not be treated under warranty coverage. This warranty is offsite in nature, the consumer will have to bring the product to the company or its authorized service centers for repair and will have to take it back at his/her own expense. No other express warranty is applicable to this product. The duration of any implied warranties, including the implied warranty of marketability or fitness for particular purpose or use is limited to the duration of express warranty herein. The company shall not be liable for the loss of use of the product inconvenience, loss or any other consequent damage arising out of the use or inability of use, of this product or for the breach of any express or implied warranty including the implied warranty of market ability or merchantability or fitness applicable to this product. All disputes are subject to Delhi jurisdiction only.

GOING ON VACATIONS

- Must put the PCU ON/OFF switch in off Position.
- Must put the HOME AWAY switch in enable position.

Instrument Description:-

Eastman Solar Smart Max Series External Battery PCU

• Two battery (24V) systems	
SOLAR SMART MAX 3600 Recommended Ah (150Ah & Above)	
• Four battery (48V) systems	
SOLAR SMART MAX 4100 Recommended Ah (180Ah & Above)	
SOLAR SMART MAX 6100 Recommended Ah (200Ah & Above)	

WARRANTY ACCEPTANCE CRITERIA

Head	Sub-Model	Action		Result
		Yes	No	
Documents	Warranty Card /Tax Invoice (Mentioned Customer details , Product Serial Number , Date of Sale etc.)	✓		Accepted
	In case of Non-Availability of Warranty Card/Invoice and proof of Purchase then warranty will be considered from the month of manufacturing.	✓		Accepted
	Product Out of Warranty as per Warranty Documents.		✓	Rejected
Physical/Technical	Product Damage i.e. Cabinet, cables, input terminal, Battery connector etc.		✓	Rejected
	Product Serial Number sticker damage or Missing.		✓	Rejected
	Mechanical or electric damages resulting from incorrect installation.		✓	Rejected
	Damages caused by acts of God, floods, fires, lighting or other natural disasters, wars, unexpected events, inappropriate voltage, defective supply materials or other external factors.		✓	Rejected
	Product outside repair, reconfiguration, modifications and adjustments.		✓	Rejected
	Damages due to the user's fault or lack of knowledge.		✓	Rejected
	Product has damaged due to any types of liquid like as Water, Acid, chemical etc.		✓	Rejected
	If Product's PCB is Physical good and not burnt then it will be considered in warranty.	✓		Accepted
	Electronic part failure in normal conditions then It will be covered in warranty.	✓		Accepted
	Product Failure due to Protection failure i.e. Overload Protection, charging failure etc.	✓		Accepted
	Product use in wrong application.		✓	Rejected

Model No:

Serial No:

Name of Purchaser:

Address:

Date of Purchase:

Dealer's Name:



For Complaint Registration Scan QR Code

MADE IN INDIA

Eastman Assurance

1800 419 8610

Eastman Auto & Power Ltd.

Corporate Office : 572, Udyog Vihar, Phase-V, Gurugram, Haryana - 122016, India | E-mail - support@eaplworld.com

+91-124-4682650 78728 78728

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A GREEN INNOVATION FROM EASTMAN GROUP

60 MONTHS WARRANTY*



SOLAR SMARTMAX Series

USER MANUAL

www.eaplworld.com

ABOUT THIS MANUAL

Purpose

This manual describes the assembly, installation, operation and troubleshooting of this unit. Please read this manual carefully before installations and operations. Keep this manual for future reference.

Scope

This manual provides safety and installation guidelines as well as information on tools and wiring.

SAFETY INSTRUCTIONS

WARNING: This chapter contains important safety and operating instructions. Read and keep this manual for future reference.

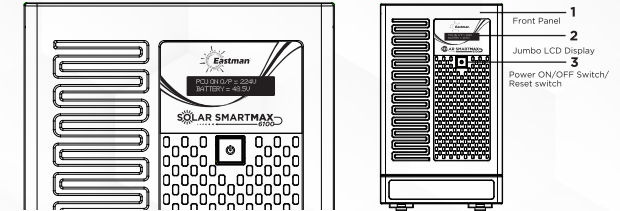
- Before using the unit, read all instructions and cautionary markings on the unit, the batteries,solar panels and all appropriate sections of this manual. Other types of batteries may burst, causing personal injury and damage.
- Do not disassemble the unit. Take it to a qualified service center when service or repair is required. Incorrect re-assembly may result in a risk of electric shock or fire.
- To reduce risk of electric shock, disconnect all wirings before attempting any maintenance or cleaning. Turning off the unit will not reduce this risk at the time of maintenance or cleaning.
- CAUTION** – Only qualified people can install this device with battery & solar panels.
- For optimum operation of this PCU, please follow required spec to select appropriate cable size. It's very important to correctly operate this PCU.
- Be very cautious when working with metal tools on or around batteries. A potential risk exists to drop a tool to spark or short circuit batteries or other electrical parts and could cause an explosion.
- Please strictly follow the installation procedure when you want to disconnect AC , Solar or DC terminals. Please refer to the INSTALLATION section of this manual for the details.
- Fuses (40A) are provided as over-current protection.
- GROUNDING INSTRUCTIONS** - This PCU should be connected to a permanent grounded wiring system. Make sure to comply with local requirements and regulations to install this inverter.
- NEVER cause AC output and DC input short circuited. Do NOT connect to the mains when DC input short circuits.
- Never connect solar panels to battery terminals directly.
- Warning!! Only qualified service persons are able to service this device. If errors still persist after following the troubleshooting table, please send this inverter/charger back to the local dealer or service center for maintenance.

FEATURES

- State of art technology using High End DSPIC for optimized performance and reliability.
- High-end DSPIC based High Resolution PWM technology using MOSFET: High-end DSPIC used where the internal digital frequency control is at a speed of as high as100 MIPS (Million instruction per second).
- ACEC (Analog Compare error correction) technology based PWM for the generation of pure SINE WAVE: Using ACEC technology the PWM is corrected.
- Boost based PWM controlled intelligent multistage Battery Management charging technology, the input power factor maintains 0.85 which reduces the electricity bills.
- Wiring fault, overload, short circuit, battery low, reverse phase, reverse battery, solar reverse, solar low, solar high protections.
- ASC (Assessing system configuration) software technique is being incorporated. The software verifies all the critical system parameters before the system starts, & keep on assessing the system parameter while the system is running.
- Deliver unmatched noise free performance.
- Narrow MODE: Very useful with computer type of application, music studio or medical equipment etc.
- CBT (Control Battery Temperature) technique is our innovation, with the help of this technique the battery temperature never exceeds the standard limit of the temperature inside the battery.
- REPLICATES MAINS: Ensure same power quality as from mains with a crest factor of as high as 3:1.
- Customer friendly Jumbo LCD display.
- PLPO: Peak Load Peak Output ensures high peak load handling capacity.
- Generator compatible.
- Static changeover by using SCR module improves life and reliability and reduces the change over time, change over time is 10ms.

- Special output with TDR (Time Delay Relay) for smooth running of compressor type load like Air Conditioner, Deep freezers etc.
- Solar Tracking efficiency above 98%.
- All over system Tested by Chroma Simulator.
- Fast response MPPT charge controller.
- Power generation calculation.
- Three working modes:- Solar, Intelligent, Hybrid.
- Three battery selections:- Tubular, Lithium, Flat.
- Lithium compatible.
- Revive battery by Solar & Grid.
- Isolated GPRS/GSM, Bluetooth, RS485, Online monitoring.

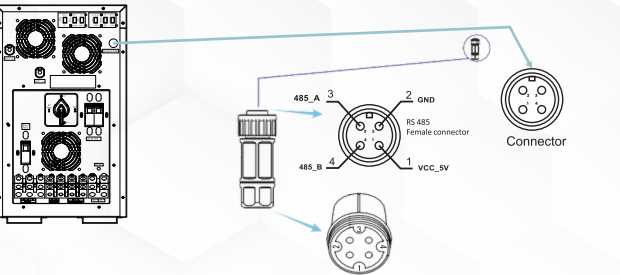
FRONT PANEL



I. POWER SWITCH INDICATION
Indicate Power Switch is ON or OFF.
II. POWER ON/OFF AND RESET SWITCH
i) When switch is OFF backup mode is inactive, Mains bypass and Charging is active. Buzzer is muted.
ii) Switch ON : All functions of product works.
iii) Switch Reset: Reset the system from all fault conditions

III. Jumbo LCD Display Indications	
Welcome Message	“Welcome to Solar Smart Max” Initialising....Solar Smart Max Initialising....S/W Version 1.1
PCU ON	Battery Volt=....V CHG Current=....A PV Volt=....V PV Current=....A O/P Volt=....V O/P Freq.=....Hz PV Pow=..... Load%=.... Total Power GEN=...WH Total Power GEN=....KWH Total Power GEN=...MWH Total Power GEN=....Unit
Mains ON	Grid Volt=....V Grid Freq.=....Hz O/P Volt=....V O/P Freq.=....Hz
Battery Indication	Battery High • Plz Check Battery Low Battery • Plz Reduce Load
Overload Indication	Overload • Plz Reduce Load
Fault Indication	i. Short Circuit • Plz Check Load ii. Over Temp. • Plz Check Venti. iii. MCB Trip • Plz Reset MCB iv. PV Power High • Plz Reduce Panel v. PV Voltage Low vi. PV Voltage High vii. Grid High viii. Grid Low

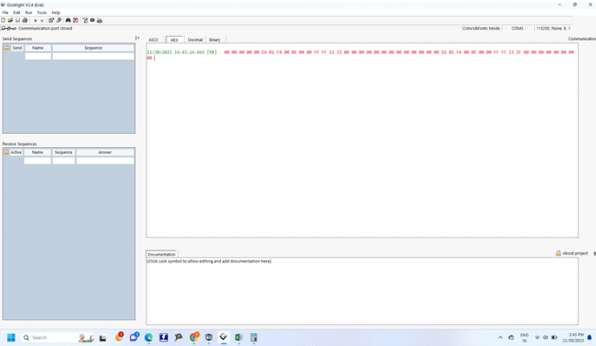
Battery Mode	• Li-ion • Tubular • Flat
Mode	• Mode Intelligent • Mode Hybrid • Mode Solar
Backup Mode ON	Grid Absent
Backup Mode OFF	Grid Bypass



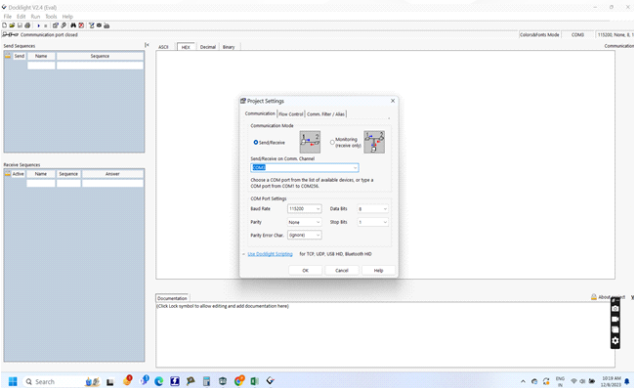
Note: If use GPRS & GSM module then connect GPRS & GSM module with this connector.

RS485 connector

RS485 DATA FORMAT			
Parameters	HEX Value	Decimal Value	Actual Value
Start Point Data	FFFF		
Battery Voltage	1333	4915	49.15V
Battery Current	00C8	200	20.0Amp
Panel Volt	00B4	180	180v
Panel Current	12C	300	30.0Amp
PV Power	1388	5000	5000W
AC Input Volt	E6	230	230V
AC Input Freq	1F4	500	50.0Hz
AC Output Volt	E7	231	231V
AC Output Freq	1F9	505	50.5Hz
Load Percentage	34	52	52%
PV Energy	101		
PV Energy	210		



Fault Indication	
1	Battery Low
2	Battery Hi
3	Over temp Inverter
4	Over temp Solar
5	Over Load
6	Short ckt
7	Panel Hi
8	Panel Low
9	PV Power Hi
10	MCB Trip
11	Mains Hi
12	Mains Low
13	Input Freq OUT



TECHNICAL SPECIFICATIONS

Input Voltage (Wide Mode)	120V-280V ± 10V		
Input Voltage (Narrow Mode)	180V-260V± 10V		
Output Voltage on Mains Mode	Same as input		
Output voltage on Backup Mode	220V + 10%, -15%		
Output frequency on Backup Mode	50Hz ± 0.1Hz		
Switching from mains to Backup and from Backup to Mains	Automatic		
Output Waveform on Mains mode	Same as input		
Output Waveform on Backup mode	Pure Sine Wave		
Battery Charging Current	Constant charging approx 10% of the rated battery current in Ah		
Charger	CCCV		
PCU overload/PCU Short Circuit	110%/150%		
PCU Transfer Time	<=15msec		
Technology	DSPIC BASED DESIGN		
Auto Reset Feature	Yes		
Operating Temperature	0°C to +45°C		
Voc RANGE	3600	4100	6100
	30V-100V	75V - 160V	75V - 160V
MAX PV Power	2500W	3000W	5000W
Solar Current MAX.	72A	40AMP	66AMP
Type of Charge	MPPT WITH ISOLATED GATE DRIVE		

*NOTE:-

- Power factor may vary depending upon the load.
- Because of a policy of continuous product improvement, specification are subject to change without notice.

SELECTION SWITCH FUNCTION -

- BATTERY TYPE -TUB/FLAT/ LITHIUM-** Select the switch according to type of battery.
TUBULAR- Keep the switch in up side Tubular Battery selected.
LITHIUM- If we keep switch down side, then Lithium Battery will be selected.
FLAT- If the battery switch up & down 3 times within 10 seconds and keep in up side and beep will ring then Flat battery will be selected.
- HOME AWAY SWITCH** - Whenever a user wants to shutdown the system permanently for some days without disconnecting the battery wire, select the switch enable for complete shutdown.it will protect the battery from discharging (HOME AWAY). But the battery will start charging by applying a grid or Solar or Both.
- CHARGING CURRENT -STD/HIGH** - Select the current according to your Battery Ah.
- GRID CHARGING ENABLE / DISABLE-** If the user wants to charge the battery only from the solar then disable grid charging.
- INPUT VOLTAGE-WIDE/NARROW** - Select the narrow mode for running IT load like a computer & for running normal load, like light,TV, fan, freeze etc. select the wide mode or select the mode according to the specification of the load that user wants run on system.
- MODE INTELLIGENT / HYBRID/SOLAR MODE-** select the mode according to availability of the grid and solar supply. system will give priority in-
INTELLIGENT MODE- Solar - Grid.
Keep the switch in up side Intelligent Mode selected.
HYBRID MODE- Grid - Solar.
If we keep switch down side, then Hybrid Mode will be selected.
SOLAR MODE- Solar only.
If the mode switch up & down 3 times within 10 seconds and keep in upside and beep will ring then Solar Mode will be selected.

SOLAR SMART MAX 6100

- DC Fan 1
- DC Fan 2
- Negative(-) Battery Lead
- Positive(+) Battery Lead
- Bypass Switch(1.Bypass 2.PCU/UPS)
- Input MCB
- Mains Input Terminal
- Output Terminal
- TDR Output
- Solar Terminal
- DC Fan 3
- Solar MCB
- RS485 Port/ANTENNA
- Grid Charging(Enable/Disable)
- Input Voltage(Wide/Narrow)
- Negative(-) Battery Lead
- Positive(+) Battery Lead
- Mode(Intelligent/Hybrid/Solar)
- Charging Current(STD/High)
- Battery(Lithium/Tubular/Flat)
- SPV Rev.

Caution: Proper selection of switch position is recommended based on the battery manufacturers specifications, for proper backup and also to avoid any damage to the battery due to wrong selection.

TROUBLESHOOTING-

Problem	Possible Caused/ Action Suggested
1. Main supply is normal but:- a) PCU is working on battery.	a) Check the AC mains presence in the mains AC wall terminal. b) Check the AC mains input voltage , it should be in the range of 120V to 280Vac when selected mode is wide & 180V to 260Vac in narrow mode.
b) No output from PCU	Check the MCB at the rear panel. If tripped, up the MCB (reset) to switch on the mains, if again tripped then call the authorized service engineer.
2. PCU trips frequently in Narrow mode.	The load might be more. Reduce the load and reset the PCU.

